

Shangyu Andrew Xu

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EDUCATION

Yale University

Aug 2023 – May 2027 (Expected)

B.S./M.S. in Computer Science, B.S. in Mathematics (Intensive)

GPA: 3.97/4.0

Coursework

ML (Grad-level): Natural Language Processing, Intermediate ML, Probabilistic ML, Deep Learning on Graphs

Math: Honors Linear Algebra, Honors Analysis, Abstract Algebra, Vector Analysis, Discrete Math

CS: Systems Programming, Spectral Graph Theory, Algorithms

Fall 2025: Probability Theory, Optimization and Computation, Building AI Infra Systems

Activities & Awards: MathCounts Outreach, 5x AIME Qualifier, USAPhO Qualifier, USNCO Qualifier

EXPERIENCE

Machine Learning Researcher

Aug 2024 – Present

Natural Language Processing (NLP) Lab, Yale University

New Haven, CT

- Designing LLM-based agents for scientific innovation and paper writing under Dr. Arman Cohan
- Benchmarking 8 frontier LLMs for multi-step graduate-level physics problems and implementing agentic system for LLMs to automate complex proof generation under Dr. John Sous

Head Teaching Assistant

Aug 2024 – May 2025

CS223: Data Structures and Programming Techniques, Yale University

New Haven, CT

- Delivered weekly section material on data structures and memory management in C for a 200-student class
- Developed 350+ public and private test cases for 6 brand-new coding projects in Fall 2024

Machine Learning Engineer Intern

May 2024 – Aug 2024

ActualFood

Remote

- Developed a full-stack iOS application in Swift, integrating a real-time CNN model for gesture recognition
- Designed and trained a CNN in PyTorch (inspired by MobileNetV4) for real-time hand gesture classification
- Quantized the CNN, reducing on-device inference time by 30% and model size by 50% on iPad devices

Business Manager

May 2024 – May 2025

Doox of Yale A Cappella

New Haven, CT

- Organized 70+ performances generating \$50k+ in revenue and 40 days of tour for 15 members across 3 continents
- Recorded 19 singers in a studio and edited, mixed, and mastered tracks with professionals for a 14 track [album](#)
- Leading the incorporation of Doox of Yale as a 501(c)(3) non-profit organization

PUBLICATIONS

Centralized Selection with Preferences in the Presence of Biases | *ICML 2024*

Nov 2023 – Jul 2024

[Paper](#) | [Slides](#) | [Code](#)

- Co-first author and presenter; authored with Dr. Nisheeth Vishnoi, Dr. Elisa Celis, and Dr. Amit Kumar
- Researched the mathematical optimization of a centralized selection algorithm under unknown bias factors
- Independently created 40+ simulations testing the robustness of core theorem under non-ideal conditions

PROJECTS

Priority-Guided Masking for Diffusion Language Models | *PyTorch, HPC*

Mar 2025 - May 2025

[Paper](#) | [Slides](#) | [Code](#)

- Final project for CPSC 477: Natural Language Processing, winning the Best Project award in a graduate class
- Fine-tuned LLaDA-8B, a state-of-the-art diffusion language model, to prioritize the generation of important tokens first, resulting in a model that is able to perform better in less inference iterations

TECHNICAL SKILLS

Languages: Python, C, Rust, C++, Swift

Machine Learning Tools: PyTorch, TensorFlow, Hugging Face, Scikit-learn, NumPy, Pandas, Matplotlib

Developer Tools: Git, VS Code, Xcode, Linux/Unix, SLURM/HPC